

OSSP SKILL

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To Apply Escape Sequences, Store Command History, Navigate Previous Commands, Navigate Next Commands, Update Input Buffer, Test Recall Functionality.

To Allocate Buffers Dynamically, Resize Arrays, Prevent Buffer Overflow, Manage Linked Lists, Release Memory Correctly, Verify with Valgrind

(a)Code

```
root@unknown
GNU nano 8.7.1 history.c *
#include <stdio.h>
#include <string.h>

int main() {
    char history[5][100];
    char command[100];
    int count = 0;

    while (count < 5) {
        printf("myshell> ");
        fgets(command, sizeof(command), stdin);

        command[strcspn(command, "\n")] = '\0';

        if (strcmp(command, "exit") == 0)
            break;

        strcpy(history[count], command);
        count++;
    }

    printf("\nCommand History:\n");

    for (int i = 0; i < count; i++) {
        printf("%d: %s\n", i + 1, history[i]);
    }

    return 0;
}
```

OUTPUT SCREENSHOT:

```
root@unknown:~# nano history.c
root@unknown:~# gcc history.c -o shell
cc1: fatal error: history.c: no such file or directory
compilation terminated.
root@unknown:~# gcc history.c -o shell
root@unknown:~# ./shell
myshell> ls
myshell> pwd
myshell> date
myshell> whoami
myshell> exit

Command History:
1: ls
2: pwd
3: date
4: whoami
root@unknown:~#
```

(b)Code:

```
root@unknown
GNU nano 8.7.1 dynamic.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

struct Node {
    char command[100];
    struct Node *next;
};

int main() {
    struct Node *head = NULL;
    struct Node *newNode;

    newNode = malloc(sizeof(struct Node));

    if (newNode == NULL) {
        printf("Memory allocation failed\n");
        return 1;
    }

    strcpy(newNode->command, "ls");
    newNode->next = head;
    head = newNode;

    printf("Command: %s\n", head->command);

    free(head);

    return 0;
}
```

OUTPUT SCREEN SHOT:

```
root@unknown:~# nano dynamic.c
root@unknown:~# gcc dynamic.c -o shell
root@unknown:~# ./shell
Command: ls
root@unknown:~#
```